United Nations

Our Common Future
Chapter 8: Industry: Producing More With Less

From A/42/427. Our Common Future: Report of the World Commission on Environment and Development

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1. Industry is central to the economies of modern societies and an indispensable motor of growth. It is essential to developing countries, to widen their development base and meet growing needs. And though industrialized countries are said to be moving into a post industrial, information-based era, this shift must be powered by a continuing flow of wealth from industry./1

2. Many essential human needs can be met only through goods and services provided by industry. The production of food requires increasing amounts of agrochemicals and machinery. Beyond this, the products of industry form the material basis of contemporary standards of living. Thus all nations require and rightly aspire to efficient industrial bases to meet changing needs.

3. Industry extracts materials from the natural resource base and inserts both products and pollution into the human environment. It has the power to enhance or degrade the environment; it invariably does both. (See Chapter 2 for a discussion of the concept of sustainable development within the context of industry and resource use.)

I. Industrial Growth and its Impact

4. As recently as 1950, the world manufactured only one-seventh of the goods it does today, and produced only one-third of the minerals. Industrial production grew most rapidly between 1950 and 1973, with a 7 per cent annual growth in manufacturing and a 5 per cent growth in mining. Since then growth rates have slowed, to about 3 per cent yearly between 1973 and 1985 in manufacturing and virtually zero growth in mining./2

5. That earlier, rapid growth in production was reflected in the rising importance of manufacturing in the economies of virtually all countries. By 1982, the relative share of value added to gross domestic product by manufacturing (the 'manufacturing value added', or MVA) ranged from 19 per cent in developing countries as a whole to 21 per cent in industrialized market economies and 51 per cent of net material product in centrally planned economies. (See Table 8-1.) If the extractive industries are taken into account, the share is even higher.

Table 8-1

Share of Manufacturing Value Added in GDP,
by Economic Grouping and Income Group

	1960	1970	1980	1962
Group of Countries	per cent			
Developing Countries	14.2	16.6	19.0	19.0
Low income	11.2	13.8	15.0	15.0
Lower-middle income	11.0	13.5	16.4	16.6
Intermediate income	10.6	14.4	17.1	17.6
Upper-middle income	19.4	21.6	24.1	23.3
High income	17.2	16.2	17.2	17.9
Developed Market Economies	25.6	26.3	27.9	27.1
Centrally Planned Economies*	32.0	42.4	50.5	50.8

* Figures refer to the share of manufacturing value added (estimated) in net material product. Data are constant (1975) prices.

Source: UNIDO, World Industry: A Statistical Review 1985 (Vienna: 1966).

1. The Changing structure of World Industry

6. In recent years, the trend of the 1950s and 1960s has been reversed. Manufacturing has declined in importance relative to other sectors of the economy. In many countries, this decline has been in progress since 1973. It is most noticeable in the case of industrial market economies, but the share of MVA in GDP has also declined in nearly half the 95 developing countries surveyed by UNIDO./3 This may reflect the growing

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interaction between industry and all fields of science and technology and the increasing integration of industry and services, as well as industry's ability to produce more with less.

7. The relative importance of industry as an employer has been declining for some time in developed countries. But the shift in jobs towards the service sector has accelerated sharply over the past 15 years with the increasing adoption of new processes and technologies. Economists continue to argue over whether the advent of an information-based economy will further depress employment in industry or will expand job opportunities overall./4

8. Most developing countries started at independence with virtually no modern industry. Then during the 1960s and 1970s their industrial production, employment, and trade consistently grow faster than these sectors in developed market economies. By 1984, developing countries accounted for 11.6 per cent of world MVA (still well short of the 'Lima target' of 25 per cent adopted by UNIDO in 1975). The centrally planned economies of Eastern Europe had raised their share of world MVA from 15.2 per cent in 1963 to 24.9 per cent in 1984./5

9. The international trade in manufactured goods, which has consistently grown fatter than has world manufacturing output, is one of the factors underlying the changing geography of industrialization. Many developing nations, particularly newly industrialized countries (NICs), have shared in this growth and made spectacular progress in industrialization. Taking the Third World as a whole, exports of manufactured goods have grown steadily relative to primary exports, rising from 13.3 per cent of their total non-oil exports in 1960 to 54.7 per cent in 1982. (See Table 8-2.)

10. In general, developing-country industrial production is diversifying and moving into more capital intensive areas such as metal products, chemicals, machinery, and equipment. And heavy industries, traditionally the most polluting, have been growing in relation to light industries. At the same time, the share of industries involved in food products, and to a lesser extent in textiles and clothing, has fallen significantly.

2. Environmental Decline and Response

11. Industry and its products have an impact on the natural resource base of civilization through the entire cycle of raw materials exploration and extraction, transformation into products, energy consumption, waste generation, and the use and disposal of products by consumers. These impacts may be positive, enhancing the quality of a resource or extending its uses. Or they may be negative, as a result of process and product pollution and of depletion or degradation of resources.

12. The negative environmental impacts of industrial activity were initially perceived as localized problems of air, water, and land pollution. Industrial expansion following the Second World War took place without much awareness of the environment and brought with it a rapid rise in pollution, symbolized by the Lob Angeles smog; the proclaimed 'death' of Lake Erie; the progressive pollution of major rivers like the Meuse, Kibe, and Rhine; and chemical poisoning by mercury in Minamata. These problems have also been found in many parts of the Third World as industrial growth, urbanization, and the use of automobiles spread./6

13. Public concern grew rapidly and forced a broad debate on environment conservation and economic growth. The possibility that the process of industrial growth would run into material resource constraints became an important theme in this debate. Although non-renewable resources are by definition exhaustible, recent assessments suggest that few minerals are likely to run out in the near future.

Table 8-2

Composition of the Merchandise Trade of Developing Countries

	Exports				Imports			
	1960	1970	1980	1982	1960	1970	1980	1982
Item	(billions of dollars)							
Primary Commodities	25	45	452	369	11	17	166	166
Non-oil	17	27	107	93	8	12	79	73
Petroleum	6	18	345	277	3	5	87	92

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Manufactures	3	9	101	112	17	39	288	296
Total	27	55	553	481	28	56	454	462
Total Non-oil	20	36	208	204	25	51	367	370
	(per cent)							
Primary Commodities Including Oil	90.4	82.6	81.8	76.8	38.8	30.1	36.6	35.9
Non-oil	62.3	49.2	19.4	19.2	28.4	21.7	17.5	15.9
Petroleum	28.1	33.4	62.4	57.5	10.4	8.4	19.1	20.0
Manufactures	9.6	17.4	18.1	23.2	61.2	69.9	63.4	64.1
	Share in Non-oil Exports				Share in Non-oil Imports			
Primary Commodities (Non-oil)	86.7	73.9	51.6	45.3	32.7	23.7	71.6	19.8
Manufactures	13.3	26.1	48.4	54.7	68.3	76.3	78.4	80.2

Source: UNIDO, Industry in a Changing World (New York 1983); for 1982, WCED estimates based on UN, 1983 International Trade Statistics Yearbook, Vol. 1 (New York: 1985)

14. By the late 1960s, growing awareness and public concern led to action by governments and industry in both industrial and some developing countries. Environmental protection and resource conservation policies and programmes were established, along with agencies to administer them. Initially policies focused on regulatory measures aimed at reducing emissions. Later a range of economic instruments were considered - taxation, pollution charges, and subsidies for pollution control equipment - but only a few countries introduced them. Expenditures rose, gradually at first, reaching 1.0 per cent and as high as 2.0 per cent of GNP in some industrial countries by the late 1970s.

15. Industry also responded to these problems by developing new technologies and industrial processes designed to reduce pollution and other adverse environmental impacts. Expenditures on pollution control measures rose rapidly in some highly polluting industries; and corporations began to set up their own environmental policy and control units. Guidelines and codes of conduct were published covering safety of products and plant operations, trade practices, technology transfer, and international cooperation./7 National and international industry associations have also developed guidelines and voluntary codes of practice./8

I am one of the patients of air pollution. When Japanese economy grew very rapidly, my asthma deteriorated. I am 39 years old. I was hospitalized when I was 18 until I was 23 because of my severe asthma. I had no joy of life, no joie de vivre in those five years. I got a job and went to work but I cannot work as long a time as ordinary people. For the last 10 years I can hardly work. And when the law was enacted, the law concerning the abatement of pollution, it has given me compensation. That is my only income, from the compensation that this law provides. And if I should suffer another disease on top of asthma, I really don't know what to do.

Yoshi Suzuki
Association of Patients of Pollution and Their Families
WCED Public Hearing
Tokyo, 27 Feb 1987

16. The results were mixed, but during the decade a number of industrial countries experienced a significant improvement in environmental quality. There was a considerable roll back in air pollution in many cities and water pollution in many lakes and rivers. Certain chemicals were controlled.

17. But these achievements were limited to some industrial countries. Taking the world as a whole, fertilizer run-off and sewage discharges into rivers, lakes, and coastal waters have increased, with resulting impacts on fishing, drinking water supply, navigation, and scenic beauty. The water quality of most major rivers has not markedly improved over the years. It is, in fact, worsening in many of them, as it is in many smaller rivers. Industrialized countries still suffer from 'traditional' forms of air and land pollution. Levels of sulphur and nitrogen oxides (NOV), suspended particulates, and hydrocarbons remain high and in some cases have

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increased. Air pollution in parts of many Third World cities has risen to levels worse than anything witnessed in the industrial countries during the 1960s./9

18. It is becoming increasingly clear that the sources and causes of pollution are far more diffuse, complex, and interrelated - and the effects of pollution more widespread, cumulative, and chronic - than hitherto believed. Pollution problems that were once local are now regional or even global in scale. Contamination of soils, ground-water, and people by agrochemicals is widening and chemical pollution has spread to every corner of the planet. The incidence of major accidents involving toxic chemicals has grown. Discoveries of hazardous waste disposal sites - at Love Canal in the United States, for example, and at Lekkerkek in the Netherlands, Vac in Hungary, and Georgswerder in the Federal Republic of Germany - have drawn attention to another serious problem.

> It is absolutely clear now that the present scale and rate of development of the productive forces require a different approach to the questions connected with environmental protection and rational utilization of natural resources. This is a task of immense economic and social significance. For actually it is a question of people's health and a caring approach to the national wealth of each country. Moreover, it is also a question of the future. And on the solution depends the conditions in which the coming generations will live.
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> A. P. Semyonov
> Central Council of Trade Unions
> WCED Public Hearing
> Moscow, 8 Dec 1986

19. In the light of this and the growth trends projected through the next century, it is evident that measures to reduce, control, and prevent industrial pollution will need to be greatly strengthened. If they are not, pollution damage to human health could become intolerable in certain cities and threats to property and ecosystems will continue to grow. Fortunately, the past two decades of environmental action have provided governments and industry with the policy experience and the technological means to achieve more sustainable patterns of industrial development.

20. At the beginning of the 1970s, both governments and industry were deeply worried about the costs of proposed environmental measures. Some felt that they would depress investment, growth, jobs, competitiveness, and trade, while driving up inflation. Such fears proved misplaced. A 1984 survey by OECD of assessments undertaken in a number of industrial countries concluded that expenditures on environmental measures over the past two decades had a positive short term effect on growth and employment as the increased demand they generated raised the output of economies operating at less than full capacity. The benefits, including health, property, and ecosystem damages avoided, have been significant. More important, these benefits have generally exceeded costs./10

21. Costs and benefits have naturally varied among industries.One method of estimating the cost of pollution abatement in industry compares expenditures on new plants and equipment that have pollution control facilities to hypothetical expenditures on new plants without such features. Studies using this comparison in the United States found that pollution abatement expenditures foe new plant and equipment for all manufacturing industries in that country in 1984 amounted to $4.53 billion, or 3.3 per cent of total new expenditures. The chemical industry spent $580 million (3.8 per cent) on such equipment./11 Similar studies in the Japanese steel industry found that new investment in pollution control equipment reached as high as 21.3 per cent of total investment in 1976 and even today remains around 5 per cent./12

22. Firms involved in food processing, iron and steel, non-ferrous metals, automobiles, pulp and paper, chemicals, and electric power generation - all major polluters have borne a high proportion of the total pollution control investment by industry. Such costs provided a strong incentive for many of these industries to develop a broad range of new processes and cleaner and more efficient products and technologies. In fact, some firms that a decade ago established teams to research and develop innovative technologies to meet new environmental standards are today among the most competitive in their fields, nationally and internationally.

23. Waste recycling and reuse have become accepted practices in many industrial sectors. In some industrialized countries technologies to scrub sulphur and nitrogen compounds from smokestack gases made remarkable advances in a relatively short time. New combustion techniques simultaneously raise combustion

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efficiency and reduce pollutant emissions./13 Innovative products and process technologies are also currently under development that promise energy- and resource-efficient modes of production, reducing pollution and minimizing risks of health hazards and accidents.

24. Pollution control has become a thriving branch of industry in its own right in several industrialized countries. High-pollution industries such as iron and steel, other metals, chemicals, and energy production have often led in expanding into the fields of pollution control equipment, detoxification and waste disposal technology, measurement instruments, and monitoring systems. Not only have these industries become more efficient and competitive, but many have also found new opportunities for investment, sales, and exports. Looking to the future, a growing market for pollution control systems, equipment, and services is expected in practically all industrialized countries, including NICs.

II. Sustainable Industrial Development in a Global Context

25. If industrial development is to be sustainable over the long term, it will have to change radically in terms of the quality of that development, particularly in industrialized countries. But this is not to suggest that industrialization has reached a quantitative limit, particularly in developing countries. Even today, according to UNIDO, world industrial output would have to be increased by a factor of 2.6 if consumption of manufactured goods in developing countries were to be raised to current industrial country levels./14 Given expected population growth, a five- to tenfold increase in world industrial output can be anticipated by the time world population stabilizes sometime in the next century. Such growth has serious implications for the future of the world's ecosystems and its natural resource base.

> Our ecological movement is not against industry, but we must think of the social function of industries and that pollution and progress are not the same thing. Pollution is not the synonym of progress and therefore time has come for new development concepts to come up. Pollution should not be a synonym of progress because we know that pollution is controlled and when you do not control pollution you are transferring this pollution to the community of the whole.
>
> Fabio Feldman
> Lawyer for Victims of Cubatao
> WCED Public Hearing
> Sao Paulo, 28-29 Oct 1985

26. In general, industries and industrial operations should be encouraged that are more efficient in terms of resource use, that generate less pollution and waste, that are based on the use of renewable rather than non renewable resources, and that minimize irreversible adverse impacts on human health and the environment.

1. Industrialization in the Third World

27. Growing populations and high proportions of young people in the Third World are leading to large increases in the labour force. Agriculture cannot absorb them. Industry must provide these expanding societies not only with employment but with products and services. They will experience massive increases in the production of basic consumer goods and a concomitant build-up of industrial infrastructure - iron and steel, paper, chemicals, building materials, and transportation. All this implies considerable increase in energy and raw material use, industrial hazards and wastes, accidents, and resource depletion.

28. The problems and prospects for industrial development vary among the countries of the Third World, which differ greatly in size and resources. There are some large countries with abundant natural resources and a substantial domestic market that provide a base for wide-ranging industrial development. Smaller, resource-rich countries are trying to build up an export-oriented processing industry. Several developing countries have based much of their industrial development on export industries in garments, consumer electronics, and light engineering. In many countries, however, industrial development is restricted to a few consumer-goods industries that cater to relatively small domestic markets.

29. The developing countries' share in world production of iron and steel rose from 3.6 per cent in 1955 to 17.3 per cent in 1984, when four countries - Brazil, China, India, and the Republic of Korea - produced more than 10 million tons of steel each, as much as in many medium-sized industrialized countries./15 At the same time that this industry is contracting in many developed countries, it is expected to expand by 38 million tons between

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1982 and 1990 in the developing world. Latin America is projected to account for 41 per cent of this rise, Southeast Asia for 36 per cent, the Middle East for 20 per cent, and Africa for 1.3 per cent./16

30. Many developing countries still depend heavily on their exports of minerals and other commodities, mostly in unprocessed or only intermediately processed forms. In the case of several major minerals such as aluminium and nickel, a few transnational corporations control the whole industry, from mining through final processing./17 Some countries have been moderately successful in increasing the share of refined products in their exports. Yet most of these 'manufactured' goods are processed further in the industrial country that imports them. Thus in 1980, only 39 per cent of all Third World exports of manufactured goods were ready for final use, while 43 per cent of its total exports were unprocessed./18 This ratio should improve as developing nations move into the further stages of processing. These improvements should be speeded up.

31. The expected growth in basic industries foreshadows rapid increases in pollution and resource degradation unless developing countries take great care to control pollution and waste, to increase recycling and reuse, and to minimize hazardous wastes. These countries do not have the resources to industrialize now and repair the damage later; nor will they have the time, given the rapid pace of technological progress. They can profit from the improvements in resource and environmental management being achieved in industrialized countries, and so avoid the need for expensive clean-ups. Such technologies can also help them reduce ultimate costs and stretch scarce resources. And they can learn from the mistakes of developed countries,

32. Economies of scale are no longer always the primary consideration. New technologies in communications, information, and process control allow the establishment of small-scale, decentralized, widely dispersed industries, thus reducing levels of pollution and other impacts on the local environment. There may, however, be trade-offs to be made: small-scale raw material processing, for example, is often labour-intensive and widely dispersed but intensive in the use of energy. Such dispersed industries could relieve big cities of some of their population and pollution pressures. They could provide non-farming jobs in the countryside, produce consumer goods that cater to local markets, and help spread environmentally sound technologies.

2. Use of Energy and Raw Materials

33. Industrial growth is widely seen as inevitably accompanied by corresponding increases in energy and raw material consumption. In the past two decades, however, this pattern appears to have fundamentally changed. As growth has continued in the developed market economies, the demand for many basic materials, including energy and water, has levelled off; in some cases, it has actually declined in absolute terms.

34. Energy consumption per unit of GDP in OECD countries has been dropping at a rate of 1-3 per cent every year since the late 1960s. Between 1973 and 1983, these nations improved energy efficiency by 1.7 per cent annually./19 Industrial water consumption per unit of production has also declined. Older pulp and paper mills typically used about 160 cubic metres of water per ton of pulp; those built during the 1970s, however, used only 70. With advanced techniques that keep water circulating within a closed system, and with proper staff training, use rates could be lowered to 20-30 cubic metres per ton of pulp./20

35. An integrated steel mill uses about 80-200 tons of water for every ton of crude steel. However, since only about 3 tons of water per ton of crude steel are lost, mostly by evaporation, recycling can greatly reduce consumption./21 Closed water circulation systems are not unique to the steel industry or to developed market economies. Between 1975 and 1980, the chemical industry's output in the USSR increased by 76 per cent, but the total consumption of fresh water remained at the 1975 level./22 And between 1981 and 1986, Soviet industrial output increased by 25 per cent but industrial water consumption remained constant./23

36. Declines in consumption of other raw materials began much earlier. In fact, the amount of raw materials needed for a given unit of economic output has been dropping over this entire century, except in wartime, for practically all non-agricultural commodities./24 A recent study of consumption trends of seven basic materials in the United States bears this out,/25 as do studies in Japan. Japan used only 60 per cent as much raw materials for every unit of industrial production in 1984 as it used in 1973./26 These efficiency trends do not result from a decline in manufacturing in favour of service industries, for over these periods the output of the manufacturing sector continued to grow The productivity and efficiency of resource use are constantly improving, and industrial production is steadily switching away from heavily material intensive products and processes.

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37. The two oil price hikes of the 1970s shocked many countries into saving money by promoting conservation measures, switching to other fuels, and raising overall energy efficiency. These events demonstrated the importance of energy pricing policies that take into account their current stock, depletion rates, availability of substitutes, and any unavoidable environmental damage associated with their extraction or processing. (See Chapter 7.) They also indicated the potential of similar pricing policies for other raw materials.

> I think there must be a persistent push, a persistent effort towards establishing some kind of international code for areas of technologies having high environmental risks. At the moment not many in Indonesia would be considered as very knowledgeable industries. We need also this kind of thing in order to guarantee some kind of minimum safety for countries like ours to develop within the context of international economic relations.
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> Speaker from the floor
> WCED Public Hearing
> Jakarta, 26 March 1985

38. Some have referred to these processes as the increasing 'de-materialization' of society and the world economy. Yet even the most industrially advanced economies still depend on a continued supply of basic manufactured goods. Whether made domestically or imported, their production will continue to require large amounts of raw materials and energy, even if developing countries progress rapidly in the adoption of resource-efficient technologies. To sustain production momentum on a global level, therefore, policies that inject resource efficiency considerations into economic, trade, and other related policy domains are urgently needed, particularly in industrial countries, along with strict observance of environmental norms, regulations, and standards.

3. Promises and Risks of New Technologies

39. Technology will continue to change the social, cultural, and economic fabric of nations and the world community. With careful management, new and emerging technologies offer enormous opportunities for raising productivity and living standards, for improving health, and for conserving the natural resource base. Many will also bring new hazards, requiring an improved capacity for risk assessment and risk management. (See Chapter 12.)

40. Information technology based chiefly on advances in micro-electronics and computer science is of particular importance. Coupled with rapidly advancing means of communication, it can help improve the productivity, energy and resource efficiency, and organizational structure of industry.

41. New materials such it fine ceramics, rare metals and metal alloys, high-performance plastics, and now composites allow more flexible approaches to production. They also contribute to energy and resource conservation, as in general they require less energy to manufacture and, being lighter, contain less matter than conventional materials.

42. Biotechnology will have major implications for the environment. The products of genetic engineering could dramatically improve human and animal health. Researchers are finding new drugs, new therapies, and new ways of controlling disease vectors. Energy derived from plants could increasingly substitute for non-renewable fossil fuels. New high-yield crop varieties and those resistant to unfavourable weather conditions and pests could revolutionize agriculture. Integrated pest management will become more common. Biotechnology could also yield cleaner and more efficient alternatives to many wasteful processes and polluting products. New techniques to treat solid and liquid wastes could help solve the pressing problem of hazardous waste disposal./27

43. Advances in space technology, now the almost exclusive domain of industrial countries, also hold promise for the Third World, even for agriculture-based economies. Weather forecasting services provided through a satellite and communications network can help farmers in deciding when to plant, water, fertilize, and harvest crops. Remote sensing and satellite imagery could facilitate optimal use of the Earth's resources, permitting the monitoring and assessment of long-term trends in climatic change, marine pollution, soil erosion rates, and plant cover. (See Chapter 10.)

44. These new technologies and the Green Revolution blur the traditional distinctions between agriculture, industry, and services. And they make it possible for developments in one sector to more radically affect those

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in another. Agriculture has become virtually an 'industry' in developed countries. Agriculture-related services - especially for regional weather forecasting, storage, and transport - are becoming ever more important. New techniques of tissue culture and genetic engineering could soon generate plant strains able to fix nitrogen from the air, a development that would drastically affect the fertilizer industry, but that would also reduce the threat of pollution by agrochemicals.

45. The chemical and energy industries are moving increasingly into the seeds business, providing new seeds that meet specific local conditions and requirements - but that may also need specific fertilizers and pesticides. Here research and development, production, and marketing need to be carefully guided so as not to make the world even more dependent on a few crop varieties - or on the products of a few large transnational

46. Yet new technologies are not all intrinsically benign, nor will they have only positive impacts on the environment. The large-scale production and widespread use of new materials, for example, may create hitherto unknown health hazards (such as the use of gallium arsenate in the microchip industry.)/28 Risk research might be carried out and products manufactured where safeguards are weak or where people are unaware of the dangers. The need for caution in introducing a new technology is reinforced by the experience of the Green Revolution, which, despite formidable achievements, raises concerns over dependence on relatively few crop strains and large doses of agrochemicals. New life forms produced by genetic engineering should be carefully tested and assessed for their potential impact on health and on the maintenance of genetic diversity and ecological balance before they are introduced to the market, and thus to the environment./29

III. Strategies for Sustainable Industrial Development

47. Resource and environmental considerations must be integrated into the industrial planning and decision-making processes of government and industry. This will allow a steady reduction in the energy and resource content of future growth by increasing the efficiency of resource use, reducing waste, and encouraging resource recovery and recycling.

1. Establish Environmental Goals, Regulations, Incentives, and Standards

48. In dealing with industrial pollution and resource degradation, it is essential that industry, government, and the public have clear benchmarks. Where the workforce and financial resources permit, national governments should establish clear environmental goals and enforce environmental laws, regulations, incentives, and standards on industrial enterprises. In formulating such policies, they should give priority to public health problems associated with industrial pollution and hazardous wastes. And they must improve their environmental statistics and data base relating to industrial activities.

49. The regulations and standards should govern such matters as air and water pollution, waste management, occupational health and safety of workers, energy and resource efficiency of products or processes, and the manufacture, marketing, use, transport, and disposal of toxic substances. This should normally be done at the national level, with local governments being empowered to exceed, but not to lower, national norms. In preparing environmental regulations, it is important that flexible systems are adopted without specifying a particular process or technology and recognizing that governments differ greatly in their capacity to formulate legal standards and enforce them.

50. Regulations to control the impacts of industrial activity across national boundaries and on the international commons are also needed. Existing or future international conventions dealing with transfrontier pollution or management of shared natural resources should enshrine certain key principles:

- the responsibility of every state not to harm the health and environment of other nations,
- liability and compensation for any damage caused by transfrontier pollution, and
- equal right of access to remedial measures by all parties concerned.

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2. Make More Effective Use of Economic Instruments

51. Pollution is a form of waste, and a symptom of inefficiency in industrial production. When industries recognize pollution as a cost, they are sometimes motivated to make investments in improved products and processes to increase efficiency and hence to reduce the pollution and waste they generate, particularly when there are economic incentives to do so, it largely depends on whether such investments will increase their economic performance.

52. But there are limits to what society can expert industry operating in competition with other industries to do voluntarily. Regulations imposing uniform performance standards are essential to ensure that industry makes the investments necessary to reduce pollution and waste and to enable them to compete on an equal footing.

53. Air and water have traditionally been regarded as 'free' goods, but the enormous costs to society of past and present pollution show that they are not free. The environmental costs of economic activity are not encountered until the assimilative capacity of the environment has been exceeded. Beyond that point, they cannot be avoided. They will be paid. The policy question is how and by whom they will be paid, not whether. Basically, there are only two ways. The costs can be 'externalized' - that is, transferred to various segments of the community in the form of damage costs to human health, property, and ecosystems. Or they can 'internalized' - paid by the enterprise. The enterprise may invest in measures to prevent the damages and, if the market for its product allows, pass the costs along to the consumer. Or it may invest in measures to restore unavoidable damage replanting forests, restocking fish, rehabilitating land after mining. Or it may compensate victims of health and property damage. In these cases, too, the costs may be passed on to the consumer.

54. Enterprises may be encouraged to invest in preventive, restorative, or compensatory measures with subsidies of various kinds. Indeed, in most industrialized and many developing countries, subsidies are a common way of encouraging companies to invest in measures needed to prevent external damage. But in this case, of course, it is the taxpayer who pays, rather than the consumer of the product. Moreover, if the subsidies are large and paid to industries operating in an international market, they can lead to trade distortions and should be avoided.

55. In 1972, the member countries of OECD agreed to base their environmental policies on a Polluter Pays Principle (PPP)./30 Essentially an economic efficiency measure, PPP is intended to encourage industries to internalize environmental costs and reflect them in the prices of products. At the same time, state regulations in CMEA countries are carried out through government bodies that allow environmental concerns to be taken into account.

> We move towards attacking the sources and not the effects. But we also meet environmental questions in our markets, among our own employees and in our local environment. This definitely provides experiences that underline the need for a more complete and comprehensive thinking about the systems of which environment becomes an integral part. We also, as industry, meet the problems of international relations and environment, unfortunately very often in the way of hidden trade barriers or difficulties in cooperation between authorities.
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> Rolf Marstrander
> Director, Environment Affairs, Norsk Hydro
> WCED Public Hearing
> Oslo, 24-25 June 1985

56. In the case of OECD, the guidelines on PPP were intended to discourage subsidies that could lead to distortions in trade. Countries agreed to phase out the use of subsidies over varying periods of time. (See Chapter 3 for the application of PPP to international trade and investment.)

57. Incentives to reduce pollution can be enhanced by other measures. Energy and water pricing policies, for example, can push industries to consume less. Product redesign and technological innovations leading to safer products, more efficient processes, and recycling of raw materials can also be promoted by a more effective, integrated use of economic incentives and disincentives, such as investment tax breaks, low-interest loans, depreciation allowances, pollution or waste charges, and non-compliance fees.

58. Sometimes the way in which other policy objectives are promoted unintentionally reduces the effectiveness of environmental programmers. For example, subsidies on raw materials or water supply or energy to promote

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the development of industry in remote areas may well dilute the pressure to conserve resources. Governments should examine whether existing economic policies, instruments, or subsidies provided to various industry-based programmes and projects contribute effectively to the promotion of environmentally sound and resource efficient practices.

3. Broaden Environmental Assessments

59. An increasing number of countries require that certain major investments be subject to an environmental impact assessment. This broader environmental assessment should he applied not only to products and projects, but also to policies and programmed, especially major macroeconomic, finance, and sectoral policies that induce significant impacts on the environment.

60. Many developing countries, particularly in Asia and Latin America, have adopted systems for environmental impact assessment. But the lack of institutional capacity and skilled personnel mean that these are often conducted by outside consultants, without quality checks. In some cases, government authorities would benefit from a second opinion on the environmental documentation they receive. Interested governments should create an independent international assessment body to help developing countries, upon request, evaluate the environmental impact and sustainability of planned development projects.

4. Encourage Action by Industry

61. Industry's response to pollution and resource degradation has not been and should not be limited to compliance with regulations. It should accept a broad sense of social responsibility and ensure an awareness of environmental considerations at a levels. Towards this end, all industrial enterprises, trade associations, and labour unions should establish company wide or industry-wide policies concerning resource and environmental management, including compliance with the laws and requirements of the country in which they operate.

62. International trade associations play a major role in Betting standards and disseminating information, which must be significantly expanded. They should establish and make widely available sectoral guidelines for assessing the sustainability and potential hazards of new facilities, for developing accident contingency plans, and for selecting pollution control or waste treatment technologies. Such key industry associations as the International Chamber of Commerce and the European Council of Chemical Manufacturers' Federation that have taken important and encouraging leadership roles in dealing with environmental issues should now take a lead in addressing the broader concerns inherent in sustainable development.

63. With limited resources at their disposal, small and medium- sized industries often find themselves unable to afford the changes necessary to meet environmental regulations and product controls. Small scale businesses such as metal working, machine tools, printing, and tanning and dying are frequently among the worst offenders of environmental regulations in any country. New technologies, especially micro-electronics, already allow small industries inexpensive means to control an entire production process. Energy-saving biological systems may be well suited to the needs of small and medium sized industries for pollution control or waste disposal.

64. Small and medium scale enterprises, constituting the largest segment of industry in most nations, need information and may in some cases require financial and technical assistance from the public sector. Management and worker training can help them incorporate cleaner technologies and environmental planning into work patterns. Governments should encourage cooperative efforts smaller firms in joint research and development on environmental issues, for example, or joint use of pollution control or waste treatment facilities.

5. Increase Capacity to Deal with Industrial Hazards

65. Chemical products have greatly improved health and life expectancies; increased agricultural production; raised comfort, convenience, and the general quality of life; and expanded economic opportunities. The chemical industry is also one of the most dynamic sectors in most countries, including many developing ones. Yet this industry, together with its products, can have a particularly severe impact on the environment. It has given rise to a host of new problems both of product and process pollution. It continues to generate an increasingly wider range of products and wastes whose effects, especially long-term ones, on human health

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and the environment are not precisely known. Major accidents have taken place, and the safety record of the industry has been challenged in recent years.

66. In a world more and more dependent on chemical products and highly complex large-scale technologies, accidents with catastrophic consequences are likely to increase. Some of the heavy metals and non-metallic minerals, such as asbestos, also pose serious hazards to health and the environment. Various hazardous products and processes are already built into current systems of production and the technological structure of contemporary society, and it will be a long time before these can be replaced with less dangerous, inherently safer technologies and systems. Some highly toxic chemicals that are known to cause cancer and birth defects and have long-term genetic effects are already in the environment in significant concentrations, and may take decades to be diffused.

5.1 Chemicals

67. Chemicals represent about 10 per cent of total world trade in terms of value,/31 some 70,000-80,000 chemicals are now on the market - and hence in the environment./32 The figure is only an informed estimate because no complete inventory has been done. Some 1,000-2,000 new chemicals enter the commercial market each year, many without adequate prior testing or evaluation of effects.

68. According to a U.S. National Research Council sample of 65,725 chemicals in common use, data required for complete health hazard evaluations were available for only 10 per cent of pesticides and 18 per cent of drugs. No toxicity data existed for nearly 80 per cent of the chemicals used in commercial products and processes inventoried under the Toxic Substances Control Act./33 This situation is now beginning to change as governments move gradually from a system of post-market testing to one of pre-market testing of all new chemicals.

69. By 1986, more than 500 chemicals and chemical products had been banned altogether or had their uses severely restricted in the country of origin./34 In addition, an unknown number of chemicals are withdrawn from clearance processes every year in the light of control agency concerns, or are never submitted to national control agencies for clearance. Some of these end up on the export market.

70. In industrial countries, in an increasingly interdependent and effective system, chemical control agencies share test results and notify each other of new restrictions on chemicals. A ban or restriction in one country is thus often quickly followed by a review and appropriate action in the others.

71. Importing developing countries do not, as a rule, share in this system. Recently, some industrial countries undertook to require their industry to provide a one-time notification to importing countries of chemicals that they have formally banned or severely restricted. They agreed to provide prior notification of the proposed export/import of such chemicals and they also agreed to provide the importing country with the information that led them to ban or restrict the chemical, if it is requested. While the intent behind this system is laudable, it is difficult to see how it can work for importing countries that have no control institutions to receive the notification or professional capacity to assess the information.

72. Third World importers have no way to effectively control trade in chemicals that have been banned or severely restricted in exporting countries. Thus these countries badly need the infrastructure to assess the risks associated with chemical use. In view of the seriousness of this situation, the Commission recommends that all governments, particularly those of the major chemical-producing countries, should:

- undertake that no new chemicals be placed on international markets until the health and environmental impacts have been tested and assessed;

- reinforce on-going efforts to obtain international agreement on the selection of existing chemicals for priority testing, on criteria and procedures for their assessment, and on a system for international sharing of the tasks and the resources required;

- strictly regulate the export to developing countries of those chemicals for which authorization for domestic sale has not been sought or given, by extending requirements for prior notification and information exchange to them; and

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- support the establishment in existing regional organizations of units qualified to receive such prior notification and information, to assess it and to advise governments in the region on the risks associated with the use of these chemicals, in order to permit individual governments to weigh these risks against benefits they may perceive from importation of the chemicals.

73. Consumer awareness should be increased. Governments should encourage the establishment of information centres on chemical products used by consumers and strengthen the international networks of information exchange, assessment, and data banks now evolving in the UN and elsewhere./35 Another essential step is the adoption and enforcement of regulations on the packaging and labelling of chemicals whose use may be potentially harmful, to ensure that clear directions are provided in common local languages. Consumer unions and other non-governmental organizations should take the lead in collecting and distributing comparative risk information on ingredients in consumer products such as cleaning agents and pesticides.

> The most explosive development in the establishment of chemical and pollutive industry has come in developing countries. This is an outright danger. The last accidents are but a few of those that may come. However, we recognize that considerable responsibility tests on the trade union movement in the individual countries in pressing for influence on authorities and managements to avoid both such accidents and investments from companies that do net follow acceptable standards.
>
> Technology development has improved environment in the industrial parts of the world. The new production and information systems make it more difficult, then, for the developing countries to use cheap labour as a means to attract industry to their countries. The future for these countries does not look very bright, unless the international society takes it upon itself to affect a sharing of production technology and resources. This is politically difficult indeed.
>
> Juul Bjerke
> International Confederation of Free Trade Unions
> WCED Public Hearing
> Oslo, 24-25 June 1985

74. The chemical producer and user industries, as the source of the risks associated with chemicals and as the greatest beneficiary of their use should bear the responsibility for ensuring (and the liability for not ensuring) that their products meet the highest standards of safety, have the fewest adverse side effects on health and the environment, and are handled with appropriate care by workers and users. This will require the fullest possible disclosure of information about the properties and production processes of chemical substances and on comparative risks, not only to the regulatory authorities but also to the workers, consumers, and residents of the community in which a chemical industry operates.

5.2 Hazardous Wastes

75. Industrialized countries generate about 90 per cent of the world's hazardous wastes. Although all estimates have a wide margin of error, given considerable differences in definition of 'hazardous waste', in 1984 some 325 million to 375 million tons were generated worldwide,/36 around 5 million tons of which were in the newly industrialized and developing areas of the world./37

76. In OECD member countries alone, thousands of waste disposal sites exist, many of which are likely to require some form of remedial action. Clean-up is expensive: Estimates include $10 billion for the Federal Republic of Germany, more than $1.5 billion for the Netherlands, $20-100 billion for the United States, and at least $60 million for Denmark (in 1966 dollars)./38 A large number of potentially hazardous sites may also exist in concentrated industrial-urban areas in centrally planned economies as well as in developing countries. Some form of government intervention is required through regulatory action or financial support.

77. Waste management in developing countries suffers from a variety of problems. Frequent and heavy rains in the tropics, for instance, leach wastes into the soils under landfills or even cause them to overflow. With little or no pretreatment of wastes, this could contaminate water supplies or cause local people to be directly exposed to the wastes. Land-filling generally occurs close to industrial states that are surrounded by poor neighbourhoods or shanty towns./39 These dangers point up the need for land use planning in developing countries, and the mote urgent need to actually implement and enforce such plans.

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78. The overriding policy objective must be to reduce the amount of waste generated and to transform an increasing amount into resources for use and reuse. This will reduce the volume that otherwise must be treated or disposed of through incineration, land disposal, or dumping at sea. This is first and foremost a problem of industrialized countries. But it is also a problem in NICs and developing countries, where rapid industrialization is bringing the same severe problems of hazardous waste management.

78. The amount of wastes crossing national frontiers is increasing and is likely to continue to do so. Between 1962 and 1983, wastes transported in Western Europe for disposal in another country virtually doubled, reaching some 250,000- 425,000 tons (1-2 per cent of the total hazardous wastes generated)./40 This increase may be attributed partly to the availability of relatively low-cost, legal, land-based disposal facilities in some countries. For example, about 4,000 shipments of hazardous wastes went from the Netherlands to the German Democratic Republic in 1984. And the Federal Republic of Germany sent about 20,000 shipments to the German Democratic Republic the preceding year. International transport of wastes meant for disposal at sea, either by incineration or dumping, amounted to about 1.8 million tons in 1983./41 Small and poor countries are especially vulnerable to offshore dumping, as has occurred in the waters of the Pacific and the Caribbean.

80. Some countries have recently proposed what amounts to a commodity trade in hazardous (including radioactive) wastes. Strengthened international cooperation in this area is vitally important, and several international bodies have taken up the matter./42 An international agreement currently being developed by OECD is to be based on three important principles: equally strict controls on shipments to non member countries; prior notification to and consent from the country of final destination, whether member or non-member country; and a guarantee of existence of adequate disposal facilities in the recipient country. UNEP has drawn up extensive draft guidelines, but as of now there is no effective mechanism either to monitor or to control hazardous waste trade and dumping./43 Governments and international organizations must more actively support efforts to achieve an effective international regime to control the transfrontier movement of hazardous wastes.

5.3 Industrial Accidents

81. Accidents involving toxic chemicals and radioactive materials can occur in plants in any region. According to a survey carried out by the U.S. Environmental Protection Agency, 6,928 accidents of varying severity occurred at U.S. plants between 1980 and 1965 - an average of five a day./44

82. In 1984, liquid gas storage tanks exploded in Mexico City, killing 1,000 people and leaving thousands more homeless. Only months after the Bhopal tragedy in India, which killed over 2,000 people and injured 200,000 more, an accident at a plant in West Virginia in the United States operated by the parent company of the Bhopal facility resulted in emergency evacuation of residents and some health problems. The accidental release in 1976 of the highly toxic and mutagenic chemical dioxin at Seveso, Italy, and the ensuing saga of drums of contaminated soil being passed around Europe, also showed that in industrial countries regulations can be evaded and minimum safety standards breached.

83. In early November 1986, a fire at a warehouse of a chemicals manufacturer in Basel, Switzerland, sent toxic fumes into France and the Federal Republic of Germany and released toxic chemicals into the Rhine, causing massive fish kills and affecting the vital water supply in countries downstream, all the way to the Netherlands. Scientists investigating the Rhine agreed that it could be years before the damaged riverine ecosystems would return to their former statue./45

84. Thus incidents at Mexico City, Bhopal, Chernobyl, and Basel - all occurring within the short lifetime of this Commission raised public concern about industrial disasters. They also demonstrated the likelihood of significant increases in the frequency and magnitude of industrial accidents with catastrophic consequences.

85. These events point to the need to strengthen national capabilities and the framework for bilateral and regional cooperation. National and local governments should:

- survey hazardous industrial operations and adopt, and enforce regulations or guidelines on the safe operation of industrial plants and on the transport, handling, and disposal of hazardous materials;

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- adopt land use policies or regional development plans that would require or provide incentives to industries that have a high pollution or accident potential to locate away from population centres, and that would discourage people from moving close to plants and waste disposal sites;

- ensure that plant workers are provided with full information about the products and technologies they handle, and are given adequate training in safe operational procedures and emergency preparedness; and

- involve local governments and community residents in major siting decisions and emergency preparedness planning.

> In industry, we feel it must now be made mandatory for any firm that is potentially polluting nature through liquid gas or particle emissions to enrol their staff in short but instructive courses of environmental education. Too often firms pollute not just through accident or design but through gross ignorance by the labour involved of the destructive effect on the environment.
>
> Donald Aubrey
> Society to Overcome Pollution
> WCED Public Hearing
> Ottawa, 26-27 May 1986

86. Increasingly, the consequences of accidents may seriously affect neighbouring countries. Nations should enter into arrangements with other nations that could be seriously affected by an accident in hazardous facilities located on its territory, under which they would agree to:

- notify each other of the location and essential characteristics of existing hazardous installations, an accident in which could spill over and affect lives, property, and ecosystems in the other;

- prepare contingency plans covering potential accidents in these installations;

- provide prompt alert, full information, and mutual assistance in case of accidents;

- establish criteria for selection of sites for new hazardous facilities, which would then be subject to the above; and

- establish standards for the liability and compensation for any damage caused by transfrontier pollution.

87. Industrial accidents and their consequences are to a large extent unpredictable. In order to better identify risks, governments, international organizations, and industry itself should promote further development of technology/risk assessment methodologies, establish data banks on such assessments conducted, and make them easily available to all countries.

6. Strengthen International Efforts to Help Developing Countries

88. Pollution-intensive, resource-based industries are growing fastest in developing countries. These governments will thus have to substantially strengthen their environmental and resource management capabilities. Even where policies, laws, and regulations on the environment exist, they may not be consistently enforced. Many developing nations have begun to build up their educational and scientific infrastructure, but their technical and institutional capacity for making the most of imported or new technologies remains small. Some countries thus continue to depend on outside technical and managerial skills for the maintenance of industrial operations. For lack of capital, they often find that a new industry can only be started with the support of foreign aid, commercial loans, a direct investment, or a joint venture with a transnational corporation.

89. The importance of private investment and the key role of transnational corporations have already been highlighted. (See Chapter 3.) It is inconceivable that a successful transition to sustainable development can be achieved unless the policies and practices are reoriented around sustainable development objectives. Those external agencies that support and facilitate private investment, particularly export credit and investment insurance organizations, should also incorporate sustainable development criteria into their policies and practices.

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90. The problems of developing-country governments are compounded by the vagaries of the international economic system, such as high debts, high interest rates, and declining terms nf trade for commodities. These do not encourage hard-pressed governments to spend high proportions of their meagre resources on environmental protection and resource management. (See Chapter 3.)

91. The developing countries themselves will eventually have to bear the consequences of inappropriate industrialization, and the ultimate responsibility for ensuring the sustainability of their development rests with each government. They must define their own environmental goals and development objectives, and establish clear priorities among competing demands on their scarce resources. They will also need to search for more self-reliant means of industrial and technological development. The choices are theirs, but they will need all the assistance - technical, financial, and institutional - that the international community can muster to help them set an environmentally sound and sustainable course of development.

92. Large industrial enterprises, and transnational corporations in particular, have a special responsibility. They are repositories of scarce technical skills, and they should adopt the highest safety and health protection standards practicable and assume responsibility for safe plant and process design and for staff training. The transnational should also institute environmental and safety audits of their plants measured against standards at other subsidiaries, not just against those of other local companies, which may have less stringent requirements These audits and their follow-up should be made available to governments and other interested parties.

93. Particular care is required in dealing with toxic chemicals and hazardous wastes, and in contingency planning for accidents. The views of non-governmental organizations and the local community should be sought in planning new industrial facilities. The relevant national and local authorities must be fully informed about the properties, potentially harmful effects, and any potential risks to the community of the technology, process, or product being introduced. The necessary information should be disclosed to nearby residents in an easily understandable manner. The enterprises must cooperate with the local government, and community in contingency planning and in devising clearly defined mechanisms for relief and compensation to pollution or accident victims.

94. Many developing countries need information on the nature of industry-based resource and environmental problems, on risks associated with certain processes and products, and on standards and other measures to protect health and ensure environmental sustainability. They also need trained people to apply such information to local circumstances. International trade associations and labour unions should develop special environmental training programmes for developing countries and disseminate information on pollution control, waste minimization, and emergency preparedness plans through local chapters.

Footnotes

1/ As will be noted later in this chapter, the conventional classification of economic activities into three sectors - primary (agriculture and mining), secondary (manufacturing), and tertiary (commerce and other services) - has become increasingly ambiguous. Some economic activities cut across all three. Furthermore, the services sector has begun to occupy an important place of its own in industrialized economies. In this chapter, however, the term 'industry' will be used in the traditional sense to include mining and quarrying, manufacturing, construction, electricity, water, and gas.

2/ GATT, International Trade 1985-66 (Geneva: 1986).

3/ UNIDO, Industry in the 1980s: Structural Change and Interdependence (New York: 1985).

4/ See, for example, W.W. Leontief, The Impact of Automation (Oxford: Oxford University Press, 1986); F. Duchin, 'Automation and its Effects on Employment', in E. Collings and L. Tanner (eds.), Employment Implications of the Changing Industrial Base (New York: Ballinger Books, 1984); J. Rada, The Impact of Micro-electronics (Geneva: ILO, 1980); and D. Werneke, Microelectronics and Office Jobs (Geneva: ILO, 1983).

5/ UNIDO, Industry and Development: Global Report 1985 (New York: 1985).

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6/ WHO, Urban Air Pollution 1973-1980 (Geneva: 1984); World Resources Institute/International Institute for Environment and Development, World Resources 1986 (New York: Basic Books, 1986).

7/ The UN Commission on Transnational Corporations has been working on a comprehensive code since 1977 but the sections on environmental and consumer protection have been virtually agreed. For other examples, see FAO, 'Code of Conduct on the Distribution and Use of Pesticides', Rome, 1985; UNEP, 'Guidelines on Risk Management and Accident Prevention in the Chemical Industry', adopted in 1982; and OECD, 'Declaration of OECD Member Countries on International Investment and Multinational Enterprise', 1976. and 'Clarification of the Environmental Concerns Expressed in Paragraph 2 of the General Policies Chapter of the OECD Guidelines for Multinational Enterprises', Paris, 1985.

8/ See, for example, International Chamber of Commerce, 'Environmental Guidelines for World industry', Paris, 1976 (revised in 1981 and 1986); Hellenic Marine Environment Protection Association, 'To Save the Seas, Declaration of a Voluntary Commitment' and 'Guidelines for the Officers of HELMEPA Member Vessels', Athens, 1982; and U.S. National Agricultural Chemicals Association, 'Guidelines on Labelling Practices for Pesticide Products in Developing Areas of the World', Washington, DC, 1985.

9/ UNEP, State of the Environment 1982 (Nairobi: 1982).

10/ OECD, 'The Impact of Environmental Measures on the Rate of Economic Growth, Rate of Inflation, Productivity and International Trade', Background Papers Prepared for the International Conference on Environment and Economics, Vol. I (Paris: 1984).

11/ U.S. Department of Commerce, 'Plant and Equipment Expenditures by Business for Pollution Abatement', Survey of Current Business, February 1986.

12/ Japanese Ministry of International Trade and Industry, data compiled annually for the Industrial Structural Council, Tokyo, 1970-86.

13/ The UN Economic Commission for Europe compiles and publishes a 'compendium of low- and non-waste technologies'. A special department in the French Ministry of Environment collects and disseminates information on clean processes and technologies ('les techniques propres').

14/ UNIDO, Industry in the 1980s, op. cit.

15/ N. Namiki, 'International Redeployment of Pollution-Intensive Industries and The Role of Multinational Corporations', prepared for WCED, 1986.

16/ OECD, Developments in Steel Making Capacity in Non-OECD Market Economy Countries (Paris: 1985).

17/ Namiki, op. cit.

18/ UNIDO, Industry in a Changing World (New York: 1983).

19/ OECD, The State of the Environment 1985 (Paris: 1985).

20/ 'Industry Experience with Environmental Problem Solving', background paper prepared for the World Industry Conference on Environmental Management, organized by the International Chamber of Commerce and the UN Environmental Programme, Versailles, 14-16 November 1984.

21/ Ibid.

22/ UNEP, The World Environment 1972-1982 (Nairobi: 1982).

23/ V. Anikeev, Director of the Department on Environment and Rational Use of Natural Resources, GOSPLAN, during a visit by WCED to the GOSPLAN headquarters, Moscow, 12 December 1986.

24/ P.F. Drucker, 'The Changed World Economy', Foreign Affairs, Spring 1986.

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25/ E.D. Larson et al. 'Beyond the Era of Materials', Scientific American, June 1986.

26/ Drucker, op. cit.

27/ For a discussion of various possibilities for industrial application of biotechnology, see J. Elkington, Double Dividends? U.S. Biotechnology and Third World Development, WRI Papers, No. 2 (Washington, DC: World Resources Institute, 1986).

28/ The 1986 annual report of the Japanese Environment Agency to the Parliament dealt extensively with this topic of the potential environmental impacts and risks posed by the new technologies. Quality of the Environment in Japan 1986 (Tokyo: 1987).

29/ The U.S. Government recently announced a comprehensive regulatory policy for ensuring the safety of biotechnology research and products. See 'Coordinated Framework for Regulation of Biotechnology', Federal Register, 26 June 1986.

30/ See OECD, 'Guiding Principles Concerning International Economic Aspects of Environmental Policies', Council Recommendation C(72)128, Paris, 26 May 1972.

31/ OECD, Economic Aspects of International Chemicals Control (Paris: 1983).

32/ The Conservation Foundation, 'Chemicals Policy in the Global Environment', paper prepared for WCED, 1986.

33/ National Research Council, Toxicity Testing (Washington, DC: National Academy Press, 1984).

34/ See 'Consolidated List of Products Whose Consumption and/or Sale Have Been Banned, Withdrawn, Severely Restricted or Not Approved by Governments', compiled by the United Nations, 1st revised edition, DIESA/WP/1, 1986.

35/ Notable examples include the International Programme on Chemical Safety (UNEP/WHO/ILO), International Register of Potentially Toxic Chemicals (UNEP), International Agency for Research on Cancer (WHO), and the UN's 'Consolidated List', op. cit.

36/ H. Yakowitz, 'Global Aspects of Hazardous Waste Management', prepared for WCED, 1985; U.S. Congress, Office of Technology Assessment, Superfund Strategy (Washington DC: U.S. Government Printing Office, 1985). U.S. estimates include wastewater in very dilute form. The result is a much larger estimate of total hazardous wastes for the United States than for other countries.

37/ Some other sources quote figures as high as 34 million tons for Brazil alone, and 22 million and 13.6 million tons for Mexico and India, respectively. See H. J. Leonard, 'Hazardous Wastes: The Crisis Spreads' National Development, April 1986.

38/ Estimates quoted in an OECD Secretariat paper, Paris, 1986.

39/ UNEP, 'Transfrontier Movements of Hazardous Wastes With Regard to Developing Countries', prepared for the Working Group of Experts on Environmentally Sound Management of Hazardous Wastes, Munich, 1984.

40/ Yakowitz, op. cit.

41/ OECD, Background Papers for Conference on International Cooperation Concerning Transfrontier Movements of Hazardous Wastes, Basel, Switzerland, 26-27 March 1985.

42/ See EEC, 'Supervision and Control of Transfrontier Shipments of Hazardous Waste', Council Directive, Brussels, December 1984; OECD, Resolution of the Council C(85)100, Paris, June 1985.

43/ UNEP "Transfrontier Movements', op. cit. See also M.J. Suess and J.W. Huismans (eds.), Management of Hazardous Waste: Policy Guidelines and Code of Practice (Copenhagen: WHO Regional Office for Europe, 1983).

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44/ Preliminary findings of a study conducted for U.S. Environmental Protection Agency, 'Acute Hazardous Data Base Washington', D.C., 1985, quoted in Yakowitz, op. cit.

45/ See, for example, La Suisse, 3-9 November; Die Welt, 10 November; Die Zeit, 14 November; Der Spiegel, 17 November; International Herald Tribune, 14-16 November 1986.

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